Solutions for Advanced Graph Theory

Sam Spiro*

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Due to limited time the following are only **sketches** of full solutions, and in particular these solutions alone wouldn't necessarily constitute a solution worth full marks. I also emphasize that there may exist other (and possibly simpler) solutions these problems.

1 HW1

0.1 *(Handshaking Lemma) Prove that every graph G has $\sum_{x \in V(G)} \deg(x) = 2e(G)$ [2-].*

Consider the set of pairs $\mathcal{P} = \{(v, e) : v \in V(G),\ e \in E(G),\ v \in e\}$ and for each vertex v let $\mathcal{P}_v = \{(v, e) : e \in E(G),\ v \in e\}$. Then the $\mathcal{P}_v$ sets partition $\mathcal{P}$, giving

$$2e(G) = |\mathcal{P}| = \sum |\mathcal{P}_v| = \sum \deg(v).$$

Alternatively you could consider the set of pairs $\{(v, w) : vw \in E(G)\}$ which gives a similar argument.

0.4 *Prove that a graph is bipartite if and only if it contains no odd cycles [2-].*

If G has an odd cycle $(v_1, \ldots, v_{2\ell+1})$ and a bipartition $V_1 \cup V_2$ with, say, $v_1 \in V_1$, then one can prove inductively that $v_i \in V_j$ iff $i \equiv j \mod 2$. But this implies $v_1 v_{2\ell+1} \in E(G)$ has both vertices in V_1, a contradiction.

For the other direction, one should prove (or at least cite) (i) a graph has an odd cycle if and only if it has a closed walk of odd length, and (ii) a graph is bipartite if and only if each of its connected components is bipartite. With this, if we assume our graph has no odd cycles then you can define a bipartition on each connected component by picking an arbitrary vertex v and defining $V_1 \cup V_2$ by having $u \in V_1$ iff $\mathrm{dist}(u, v)$ is odd. One can check that this can not have, say, $uu' \in E(G)$ with $u, u' \in V_1$ as otherwise this plus the paths from u, u' to v would create an odd closed walk.

1.3a *Prove that*

$$z(m, n; s, t) \leq (t-1)^{1/s} m n^{1-1/s} + (s-1)n.$$

(Hint: if you're struggling with this, try solving the previous problem first) [2].

*Dept. of Mathematics and Statistics, sspiro@gsu.edu

Consider the set $\mathcal{P}$ of pairs (S, v) where $v \in V$ and $S \subseteq N(v)$ is a set of size s. If G avoids a $K_{s,t}$ then each of the $\binom{m}{s}$ sets S can belong to at most $t-1$ pairs, proving $|\mathcal{P}| \leq (t-1)\binom{m}{s}$. On the other hand a convexity argument shows $|\mathcal{P}| \geq n\binom{n^{-1}e(G)}{s}$, and now some algebra gives the result.

1.3b *Prove that if G is an n-vertex bipartite C_4-free graph then $e(G) \leq 2^{-3/2}n^{3/2} + o(n^{3/2})$, i.e. the lower bound we got for $\mathrm{ex}(n, C_4)$ using G_q was best possible in the setting of bipartite graphs [2-].*

Let G be such a graph with parts of sizes m and $n-m$, and (importantly) without loss of generality we may assume $m \leq n/2$. By the first part with $s = t = 2$ we have $e(G) \leq m(n-m)^{1/2} + (n-m) = m(n-m)^{1/2} + o(n^{3/2})$. One can check (using calculus, for example), that this expression is maximized in the range $0 \leq m \leq n/2$ at the value $m = n/2$, giving the bound. Note crucially that if we did not assume $m \leq n/2$ then the maximum would be at $m = 2n/3$, which would give a suboptimal bound.

1.3c *Prove that for all s, t there exists a constant $C > 0$ such that if G is an n-vertex $K_{s,t}$-free graph, then the number of edges $xy \in E(G)$ with $\deg(x) \geq Cn^{1-1/s}$ is at most $O(n)$. Find an example of a graph which has $\Theta(n)$ edges of this form (Hint: the intended proof I have in mind works with $C \approx (s+t-1)^{1/s}$) [2].*

Define an auxilliary bipartite graph B where one part U consists of all vertices of G with $\deg_G(x) \geq Cn^{1-1/s}$ and the other part V consists of a disjoint copy of $V(G)$ where we have $xy \in E(B)$ if $xy \in E(G)$ and $x \in U$. Observe that B can not contain a $K_{s,t+s}$ with the part of size s in U, since if it did then removing the at most s vertices that appear in both parts of the $K_{s,t+s}$ from the part of size $t+s$ would give a $K_{s,t}$ in G. It follows from (a) that $e(B) \leq (s+t-1)^{1/s}|U|n^{1-1/s} + (s-1)n$. On the other hand, by construction we have $e(B) \geq Cn^{1-1/s}|U|$, so if say $C = 2(s+t-1)^{1/s}|U|n^{1-1/s}$ then this implies $(s-1)n \geq (s+t-1)^{1/s}|U|n^{1-1/s}$, and hence $e(B) \leq (s+t-1)^{1/s}|U|n^{1-1/s} + (s-1)n \leq 2(s-1)n$. But $e(B)$ is exactly the number of edges of the form that we wish to bound, proving the result.

An example of a graph which this is tight is an n-vertex star.

1.4a *Prove that if G is an n-vertex graph then G contains at least $e(G) - \mathrm{ex}(n, F)$ copies of F for any graph F with at least one edge [1].*

One can prove this by induction on $\Delta := e(G) - \mathrm{ex}(n, F)$, for example, the case $\Delta = 0$ being trivial. Inductively for $\Delta > 0$, we have by definition of $\mathrm{ex}(n, F)$ that G contains some copy of F, let e be any such edge. Then inductively $G - e$ contains at least $\Delta - 1$ copies of F and these copies must be distinct from the copy of F containing e (since these copies live in $G - e$), giving the desired result.

1.4b *Prove that if G is an n-vertex graph with $e(G) \geq 100n^{3/2}$ then G contains at least $\Omega(n^{-4}e(G)^4)$ copies of C_4 (the number 100 does not matter in case you'd rather prove this result with a different constant) [2].*

For notational convenience let $d(u, v) = |N(u) \cap N(v)|$. It is not too difficult to see that

the total number of C_4's in a graph G is $\sum_{u \neq v} \binom{d(u,v)}{2}$. By convexity this is at least

$$\binom{n}{2}\left(\binom{n}{2}^{-1}\frac{\sum d(u,v)}{2}\right) \approx n^{-2}(\sum d(u,v))^2.$$

Note that $\sum d(u,v) = |\mathcal{P}|$ the set of pairs that we defined in class, and this in turn is lower bounded by roughly $n^{-1}e(G)^2$ by the convexity argument we did in class, and plugging this in gives the desired result.

2 HW2

1.8 *Verify that if G' is an n-vertex complete $(r-1)$-partite graph then $e(G') \leq e(T_{r-1}(n))$ [1+].*

Let $n_1, \ldots, n_{r-1}$ be the sizes of a complete $(r-1)$-partite graph on n vertices. Then its number of edges equals

$$\sum_{i<j} n_i n_j.$$

If there exists some i, j with say $n_i \geq 2 + n_j$, then one can consider a new sequence of integers defined by replacing n_i, n_j with $n_i - 1, n_j + 1$ and one can easily check that this strictly increases the sum above. As such, the sum is maximized when all of these integers are within 1 of each other, and this in turn is only possible if each value is equal to the floor or ceiling of $n/(r-1)$ (since in particular, some value must be at least the floor and some value must be at least the ceiling simply by averaging).

1.9a *Observe that if G is a triangle-free graph, then $\deg(x) + \deg(y) \leq v(G)$ for all $xy \in E(G)$. Use this to prove Mantel's Theorem (which is in fact the original way Mantel proved his result) [2].*

Let G be an n-vertex triangle-free graph. Then our observation above implies

$$\sum_{xy \in E(G)} \deg(x) + \deg(y) \leq ne(G).$$

On the other hand, each term $\deg(x)$ in this sum appears exactly $\deg(x)$ times, meaning the sum equals $\sum_x \deg(x)^2$. Note that Cauchy-Schwarz implies that for any sequence of n real numbers x_i that $n\sum_i x_i^2 = \sum_i 1^2 \cdot \sum_i x_i^2 \geq (\sum_i x_i)^2$. Applying this here gives $\sum \deg(x)^2 \geq n(2e(G))^2$ which exactly gives the bound that we want.

1.9b *Generalize our inductive proof of Mantel's Theorem to give an alternative proof of Turán's Theorem (which is in fact the original way that Turán proved his result).*

Instead of deleting a pair of vertices in an edge like we did for Mantel, we now delete a set of vertices forming a K_{r-1}. In this case no vertex outside the K_{r-1} can be adjacent to every vertex of this K_{r-1}, and the same sort of analysis as we did before proves the result.

1.10 *Let F denote the unique 4-vertex graph with 5 edges (i.e. the graph consisting of two triangles sharing an edge). Prove (without using* **??***) that* $\mathrm{ex}(n, F) = \lfloor n^2/4 \rfloor$ *for all* $n \geq 4$ *[2].*

We prove this by induction on n, the base cases being straightforward. If we have a graph with more than $n^2/4$ edges then we know there exists some triangle xyz. Look at $G - x - y - z$ and observe now that every vertex here has at most one neighbor in x, y, z since otherwise we create our forbidden graph. Then the same analysis we did for our previous inductive proof gives the result.

3 HW3

1.13 *Prove that for every integer* $s \geq 1$ *and real* $\varepsilon > 0$*, there exists a graph with average degree at least* $2s - \varepsilon$ *which contains no non-empty subgraph with minimum degree greater than* $s + 1$*; that is, the* $d/2$ *in* **??** *is essentially best possible [2-].*

$K_{s,t}$ with t large has average degree $\frac{2st}{s+t} = 2s - \varepsilon$, and we can guarantee min degree at least s but nothing more.

1.14 *Prove (without using* **??***) that for all* $\ell \geq 2$ *we have* $\mathrm{ex}(n, \{C_3, C_4, \ldots, C_{2\ell}\}) = O(n^{1+1/\ell})$ *(Hint: first prove the result under the additional assumption that every vertex of* G *has degree at least* $n^{1/\ell} + 1$*) [2].*

We claim that if G is a graph which has no cycles of length at most 2ℓ and minimum degree δ, then for all $0 \leq i \leq \ell$ and every vertex u there exist at least δ^i vertices v such that there exists a directed walk of length i from u to v. We prove this by induction on i, the base case $i = 0$ being trivial. Note that for each vertex v' which can be reached by a walk of length $i - 1$ there exist at least δ vertices which can be reached by a walk of length i, namely by appending any neighbor of v' to its walk. On the other hand, each vertex can be reached by at most one walk of length i as otherwise there would exist a short cycle in G. This implies that the number of vertices we can reach increases by a factor of δ, proving this claim.

For the actual problem, it suffices to prove the result under this min degree condition due to our lemma allowing us to translate from average degree d to min degree $d/2$. By our claim, any given vertex of our graph can reach more than n vertices, a contradiction to our graph having only n vertices.

2.1a *Prove that* $\mathrm{ex}(n, C_n) = \binom{n-1}{2} + 1$ *[2].*

Lower bound comes from K_{n-1} plus a leaf. For the upper bound, take an n-vertex C_n-free graph G with at least $\binom{n-1}{2} + 2$ edges. If we look at two non-adjacent vertices x, y then trivially $\deg(x) + \deg(y) \geq e(G) - \binom{n-2}{2} \geq n$ with the first inequlity using that every edge of G is either between two vertices of $G - x - y$ (of which there can be at most $\binom{n-2}{2}$) or is incident to exactly one of x, y. This together with Ore gives the result.

2.4 *Prove Pósa's Theorem [2].*

Go through our same proof of Dirac's Theorem except this time pick x_1, x_n to be non-adjacent vertices with $\deg(x_1) + \deg(x_n)$ as large as possible. If $\deg(x_1) + \deg(x_n) \geq n$ then we are done as before. Otherwise we must have, say, $\deg(x_1) := k < n/2$. Now for each of the k neighbors x_i of x_1, we must have that $x_{i-1} \not\sim x_n$, which by our choice of non-edge x_1, x_n means that $\deg(x_{i-1}) \leq k$ for all k of these vertices. This contradicts our hypothesis on Ore's Theorem.

4 HW4

10.1 *Prove that if G is connected and has diameter d, then A has at least $d+1$ distinct eigenvalues (Hint: it suffices to show that the minimum polynomial of A has large degree) [2].*

By standard linear algebra facts, if $f(x)$ is a non-zero polynomial of minimum degree such that $f(A) = 0$ then A has $\deg(f)$ distinct eigenvalues. Let f be such a polynomial and assume for contradiction that $\deg(f) = r \leq d$. Because G is connected and of diameter d, there exists a pair of vertices u, v at distance exactly r from each other. Because $A^k_{u,v}$ counts the number of walks from u to v, we have that $A^k_{u,v} = 0$ for all $0 \leq k < r$ and $A^r_{u,v} > 0$. As such, if we look at the u, v entry of $f(A)$ then this will be non-zero, notably it will be $A^r_{u,v}$ times the (non-zero) coefficient of x^r in f. This contradicts $f(A) = 0$.

10.5a *Prove that a graph is bipartite if and only if the spectrum of its adjacency matrix is symmetric about 0 [2].*

If G has an odd cycle of length $2\ell+1$, then G has a closed walk of length $2\ell+1$, meaning $\sum \lambda_i^{2\ell+1} \neq 0$ which is impossible if the spectrum of A were symmetric about 0.

On the other hand, if G has a bipartition $U \cup V$ then for every eigenvector x corresponding to λ there exists an eigenvalue x' corresponding to $-\lambda$ obtained by taking $x'_u = x_u$ for $u \in U$ and $x'_v = -x_v$ for $v \in V$. One can check that the x' vectors are orthogonal if the x vectors were, so this shows that each eigenvalue $-\lambda$ appears with the same multiplicity as that of λ.

10.5b *Prove that a connected graph is bipartite if and only if $\lambda_{\min} = -\lambda_{\max}$ [2].*

Note: there was a typo in the original problem that didn't ask for the connected condition which makes this problem somewhat more challenging than I intended.

If G is bipartite then $\lambda_{\min} = -\lambda_{\max}$ by the previous part. Now assume $\lambda_{\min} = \lambda_{\max}$ and let x be a unit eigenvector corresponding to $\lambda_{\min}$. By the raleigh quotient we have

$$-\lambda_{\min} = -x^T A x = -\sum_{uv \in E(G)} x_u x_v \leq \sum_{uv \in E(G)} |x_u x_v| \leq \max_{y : ||y|| = 2} y^T A y = \lambda_{\max}.$$

By assumption these inequalities must be equalities. This first inequality means that $x_u x_v$ is non-negative for all $uv \in E(G)$. The second equality means that the vector $|x|$ of absolute values of x is an eigenvector for G corresponding to $\lambda_{\max}$, which by well-known facts about Perron vectors (or which can be proven directly) means $|x|$ has all strictly

positive entries, i.e. x has no 0 entries. In particular, if we define U to be the set of vertices of G with $x_u > 0$ and V the set of vertices with $x_v < 0$ then this defines a bipartition for G.

10.7 *Prove that there exist two graphs G_1, G_2 with adjacency matrices A_1, A_2 such that $\sigma(A_1) = \sigma(A_2)$ and such that G_1 is connected while G_2 is not connected (Hint: there exist examples where G_1, G_2 have 5 vertices each) [2-].*

The simplest example is to take $G_1 = K_{1,4}$ and $G_2 = C_4 \cup K_1$. Their spectrums can be deduced from the previous exercise, for example, or simply by computing things by hand.

5 HW5

3.6a *Prove that $R_q(3) > 2^q$ [2-].*

We give a coloring of the edges of K_N with $N = 2^q$ as follows: identify each vertex by a binary string of length q. For two distinct binary strings u, v, we color the edge between them i if $u_j = v_j$ for $j < i$ and if $u_i \neq v_i$. Such a coloring can't have a triangle in color i since any three vertices u, v, w must have two words which agree on position i (since the only possible values are 0 and 1). Alternatively you can prove this inductively by splitting the vertex set in half, giving the coloring for $q-1$ colors within each part, and then using color q for the edges across the parts.

3.6b *Prove that $R_q(3) \leq 3 \cdot q!$, noting that this is best possible for $q = 2, 3$ [2].*

We can generalize the Erdős-Szekeres argument. To this end, if in a coloring of K_N any vertex v has a neighborhood in some color i of size at least $R_{q-1}(3)$, then either this neighborhood uses color i (in which case it plus v gives a monochromatic triangle) or it only uses $q-1$ colors, in which case it has a monochromatic triangle by construction. We conclude that each vertex v must have at most $R_{q-1}(3)$ neighbors of each color, meaning that if N has a good coloring then $N \leq 1 + q(R_{q-1}(3) - 1)$. Since such a coloring exists with $N = R_q(3) - 1$, we find inductively that $R_q(3) \leq 2 + q(R_{q-1}(3) - 1) \leq 2 + q(3(q-1)! - 1) = 3q! + 2 - q \leq 3q!$.

3.7 *For every graph F and integer q, define $R_q(F)$ to be the smallest integer N such that any q-edge coloring of K_N contains a monochromatic copy of F. Prove that if $\mathrm{ex}(n, F) = O(n^{2-\alpha})$ for some $\alpha > 0$, then $R_q(F) = O(q^{1/\alpha})$ (Hint: concretely if you assume $\mathrm{ex}(n, F) \leq Cn^{2-\alpha}$ then you should be able to prove something like $R_q(F) \leq (4Cq)^{1/\alpha}$, for example) [2-].*

Assume C is such that $\mathrm{ex}(n, F) \leq Cn^{2-\alpha}$ and consider $N > (4Cq)^{1/\alpha}$ and some q-edge coloring of K_q. By pigeonhole, some color is used at least

$$q^{-1}\binom{N}{2} \geq \frac{1}{4}q^{-1}N^2 = \frac{1}{4}q^{-1}N^{\alpha}N^{2-\alpha} > CN^{2-\alpha}$$

times. But by assumption on $\mathrm{ex}(n, F)$ this means the N-vertex graph using this popular color has a copy of F, a contradiction.

3.9c *Prove that $\hat{R}(n) = \binom{R(n)}{2}$; noting crucially that this equality holds despite us largely not understanding what $R(n)$ is (Hint: prove that any K_n-Ramsey graph must have chromatic number at least $R(n)$) [2+].*

Assume this claim was false. Consider some partition $V_1 \cup \cdots \cup V_N$ of G into $N = R(n) - 1$ independent sets. By definition of N, there exists some red-blue coloring χ of K_N avoiding monochromatic K_n. Define a coloring χ' on G by having each $v_i v_j \in E(G)$ with $v_i \in V_i$ and $v_j \in V_J$ have the same color as $\chi(ij)$ in K_N. One can check that this has no monochromatic K_n since K_N didn't have one, contradicting G being K_n-Ramsey.

Now we claim that any graph of chromatic number k must have at least $\binom{k}{2}$ edges. Indeed, if this weren't the case then there would be two parts V_i, V_j in a coloring with no edges between them, and hence we could merge these into one part and drop our chromatic number by 1.

These two claims together give the result.

6 HW6

4.3 *Prove that there exists some $C > 1$ such that every n-vertex graph G with maximum degree Δ has at least C^n proper $(\Delta+1)$-colorings. That is, G does not just have 1 proper $(\Delta+1)$-coloring but at least exponentially such colorings (Hint: the proof we have in mind yields that there are at least $2^{\frac{\Delta}{\Delta+1}n}$ such colorings, in particular giving the result with $C = 2^{2/3}$) [2+].*

We inductively prove a lower bound of $2^{\frac{\Delta}{\Delta+1}n}$ by induction on n for a given value $\Delta \geq 1$, the result being trivial for $n = 1$ since we have at least $\Delta + 1 \geq 2 \geq 2^{\frac{\Delta}{\Delta+1}}$ ways to color K_1. If G has a vertex v of degree at most $\Delta - 1$, then we can color $G - v$ and then color v in at least $2^{\frac{\Delta}{\Delta+1}(n-1)} \cdot 2 \geq 2^{\frac{\Delta}{\Delta+1}n}$ ways since v has at least 2 colors in $[\Delta+1]$ that we can give it. If G is Δ-regular then take any vertex v, color $G - v - N(v)$, then color $N(v)$, then color v. The total number of choices for this is at least $2^{\frac{\Delta}{\Delta+1}(n-\Delta-1)} \cdot 2^{\Delta} \cdot 1 = 2^{\frac{\Delta}{\Delta+1}n}$ since each neighbor of v has at most $\Delta - 1$ colors in its neighborhood when we color it since v hasn't been colored yet.

4.4 *Prove that one can equivalently define the degeneracy $d(G)$ of a graph G to be the smallest integer d such that every subgraph $G' \subseteq G$ has $\delta(G') \leq d$, i.e. has a vertex of degree at most d, [2].*

Let $n = v(G)$ and let $G_n = G$. Let v_n be a vertex of G_n of minimum degree, then take $G_{n-1} := G_n - v_n$, and iteratively let v_i be a vertex of minimum degree of G_i and define $G_{i-1} = G_i - v_i$. Observe that by definition of degeneracy, $d(G) \leq \max_i \deg_{G_i}(v_i) = \max_i \delta(G_i) \leq d$ since by definition every G_i contains a vertex of degree at most d.

For the other direction, let $G' \subseteq G$ be a subgraph with minimum degree d. It is not difficult to see then that for any ordering $v_1, \ldots, v_n$ of $V(G)$, if v_i is the largest integer such that $v_i \in V(G')$ then $|\{v_j : j < i,\ v_j \in N_G(v_i)\}| \geq \deg_{G'}(v_i) \geq d$, and since this holds for any ordering we must have $d(G) \geq d$.

4.8 *Prove that our two stated versions of the Hajnal-Szemerédi Theorem (**??** and **??**) are equivalent to each other [2-].*

First assume that we can prove the coloring version. Let G be a graph with $r|n$ and $\delta(G) \geq (1-1/r)n$. This means that the complement of the graph has maximum degree at most $n/r-1$, so by the color version we can find an equitable coloring of $\overline{G}$ using n/r colors (which is an integer since $r|n$). But because $r|n$ this means each color class is an independent set of size exactly r in $\overline{G}$ which is exactly a K_r factor in G.

Now assume we can prove the factor version, let G be a graph of max degree Δ, and let $0 \leq m \leq \Delta$ be the unique integer such that $v(G)+m$ is divisible by $\Delta+1$. Let $G' = G \cup K_m$, noting that this graph still has maximum degree at most Δ. Now look at the complement of G', it has minimum degree at least $m+n-1-\Delta = (1-(1+\Delta)/(m+n))(m+n)$. Since $(m+n)/(\Delta+1)$ is an integer which divides $m+n$, we can find a K_r factor in the complement of G'. This corresponds to a $\Delta+1$ coloring of G' where every color class has exactly the same size and by construction each color class contains at most one vertex of the K_m. Thus deleting these vertices gives an equitable coloring of G.

4.13a *Complete our proof of **??** by showing that for all t there exists some n_t such that $\chi_\ell(K_{n_t,n_t}) \geq t$ [2].*

We prove this for $n_t = \binom{2t-1}{t-1}$ via taking L to give each vertex in one part of the graph each possible subset of $[2t-1]$ of size $t-1$ and the same on the other side. One can show any proper L-coloring must use at least $t-1$ colors on each side, but then this means some vertex on the other side can't be colored while avoiding this set of colors.

7 HW7

4.16 *Given a graph G and an integer $k \geq 1$, we define the Zykov graph $Z(G,k)$ by taking k disjoint copies $G_1, \ldots, G_k$ of G, and then for each of the $v(G)^k$ sequences $\vec{x} = (x_1, \ldots, x_k) \in V(G_1) \times \cdots \times V(G_k)$ we add a new vertex $v_{\vec{x}}$ whose neighborhood equals $\{x_1, \ldots, x_k\}$. Prove that $Z(G,k)$ is triangle-free whenever G is, and that $\chi(Z(G,k)) = \chi(G)+1$ provided $k \geq \chi(G)$. As such, these graphs give another explicit family of triangle-free graphs with arbitrarily large chromatic numbers [2].*

No triangle can use two $v_{\vec{x}}$ vertices since these have no edges between them, it also can't use three vertices from the same G component since G is triangle-free, but also it can't use two vertices from distinct G components because there's no edges between them. Since $v_{\vec{x}}$ has at most one neighbor in each G component this rules out all the cases.

The upper bound on the chromatic number comes from giving a proper coloring to each copy of G and then using a new color on each of the $v_{\vec{x}}$ vertices. For the lower bound, each G component must use at least $\chi(G)$ colors, and from this one can argue that some $v_{\vec{x}}$ must be adjacent to at least $\chi(G)$ colors, meaning at least $\chi(G)+1$ colors are needed.

7.1a *Prove that every graph G contains a bipartite subgraph G' with $e(G') \geq \frac{1}{2}e(G)$ using probability.*

Let $U \subseteq V(G)$ be a random set of vertices obtained by keeping each vertex independently and with probability 1/2 and let $V = V(G) \setminus U$. Let G' be the graph consisting of every edge uv with one vertex in U and the other in V. This is a bipartite graph, and the expected number of edges it has is $e(G)/2$ (the probability a given edge lies in G' is 1/2 because conditioning on where u lands gives a 1/2 chance of v landing in the other set). There thus must exist a choice of U which gives at least $e(G)/2$ edges, proving the result.

7.2a *(Caro-Wei) Prove that if G is an n-vertex, then*

$$\alpha(G) \geq \sum_{x \in V(G)} \frac{1}{\deg(x)+1}.$$

Let $v_1, \ldots, v_n$ be a uniform random ordering of the vertex set and let I be the random set obtained by including v_i if and only if $v_j \notin N(v_i)$ for any $j < i$, ie if v_i appears in the ordering before any of its neighbors. It is not difficult to see that I is an independent set and that $\Pr[x \in I] = \frac{1}{\deg(x)+1}$, so the result follows from linearity of expectation.

7.3b *Prove that if $N \geq m$ and $e(F) \geq 2$, then G contains an F-free subgraph with at least*

$$\Omega\left(\frac{m^{1+\frac{1}{e(F)-1}}}{N^{\frac{1}{e(F)-1}}}\right)$$

edges [2].

Let $G' \subseteq G$ be obtained by keeping each edge independently and with probability $p = \frac{1}{2}(m/N)^{\frac{1}{e(F)-1}}$. In this case $\mathbb{E}[e(G')] = pm = \frac{1}{2}m^{e(F)/(e(F)-1)}N^{-1/(e(F)-1)}$ and the expected number of copies of F in G is $p^{e(F)}N = 2^{-e(F)}m^{e(F)/(e(F)-1)}N^{-1/(e(F)-1)}$. In particular, if we take G' and delete an edge from each copy of F then this gives an F-free graph whose expected number of edges is at least $\frac{1}{2}m^{e(F)/(e(F)-1)}N^{-1/(e(F)-1)} - \frac{1}{4}m^{e(F)/(e(F)-1)}N^{-1/(e(F)-1)} \geq \frac{1}{4}m^{e(F)/(e(F)-1)}N^{-1/(e(F)-1)}$, so there must exist such a graph with at least this many edges.

8 HW 8

5.3 *Use Tutte's Theorem to prove that any n-vertex graph G with n even and $\delta(G) \geq n/2$ has a perfect matching; note that your proof must somewhere use that n is even [2].*

The solution I have is a bit more involved than I thought it would be, so I'd be happy to see cleaner solutions. Anyways, let $S \subseteq V(G)$ be an arbitrary set and assume for contradiction that there is at least $|S|+1$ odd connected components in $G - S$. Note that this means one of these components has size at least $(n - |S|)/(|S|+1)$. On the other hand, every component must have size at least $\delta(G) - |S| + 1 \geq n/2 - |S| + 1$ since the neighborhood of any vertex of the component must be contained in the rest of the component and S while having size at least $n/2$. In total then we have

$$n/2 - |S| + 1 \leq \frac{n - |S|}{|S|+1}.$$

After some algebraic manipulations one can check that this inequality can only hold if either $|S| = 0$ or if $|S| > n/2$. In the latter case, because the number of components is at most $n - |S|$ this is trivially at most $|S|$ and we win. For $|S| = 0$, because $v(G)$ is even G must have at least two connected components, one of which has size at most $n/2$, but this contradicts every component having size at least $n/2 - |S| + 1 = n/2 + 1$, proving the result.

5.5 *A d-regular graph G is said to have a 1-factorization if it contains d edge disjoint perfect matchings $M_1, \ldots, M_d$ (which means every edge of G is contained in exactly one such matching). Prove that every d-regular bipartite graph has a 1-factorization [2].*

It suffices to prove that every regular bipartite graph has a perfect matching, as from there the result will follow by induction on d. Let G be a d-regular bipartite graph with bipartition $U \cup V$, noting that $d|U| = \sum_{u \in U} \deg(u) = \sum_{v \in V} \deg(v) = d|V|$ so $|U| = |V|$ (with the middle equality using that the graph is bipartite). Now consider any $X \subseteq U$. Because X is incident to exactly $d|X|$ edges, we have that some vertex of $N(X)$ must be incident to at least $d|X|/|N(X)|$ edges. If $|N(X)| < |X|$ then this implies a vertex of G has degree larger than d, a contradiction. Thus $|N(X)| \geq |X|$ and Hall's Theorem applies.

6.4 *Write an integer program whose optimum value is $\mathrm{ex}(n, K_3)$ and prove that the optimum value of its linear relaxation is asymptotically larger than $n^2/4$ [2].*

Take the integer program which has variables $x_{i,j} \in \mathbb{Z}$ for all distinct $i, j \in [n]$ defined by

$$\begin{aligned} \text{maximize} \quad & \sum_{1 \leq i < j \leq n} x_{i,j} \\ \text{subject to } & 0 \leq x_{i,j} \leq 1 \quad \forall i \neq j \\ & x_{i,j} + x_{i,k} + x_{j,k} \leq 2 \quad \forall i \neq j \neq k. \end{aligned}$$

Note that any assignment $x_{i,j}$ satisfying these conditions has $x_{i,j} \in \{0, 1\}$ for all i, j since $0 \leq x_{i,j} \leq 1$ must be an integer. Thus we can define a graph G on $[n]$ by setting $ij \in G$ if and only if $x_{i,j} = 1$, and reversing the logic above shows that G is triangle-free with number of edges exactly equal to the output of this program.

For the linear relaxation we can take $x_{i,j} = 2/3$ for all i, j to obtain a value of $2/3\binom{n}{2}$.

9 HW 9

8.4d *Prove that there exist $A \subseteq [n]$ with no 3AP such that*

$$|A| \geq n e^{-C\sqrt{\log n}},$$

for some $C > 0$ [2].

We aim to use the previous parts. Let A' be the set guaranteed by (b) for some m, d. We want to (essentially) work with $\phi(A')$, and to this end we take $m = \frac{1}{2} n^{1/d}$ to guarantee that

$\phi(A') \subseteq [n] \cup \{0\}$. This gives a set of size $\frac{n}{2d2^d n^{2/d}}$ and this is minimized when $2^d \approx n^{2/d}$, so we choose to take $d = \sqrt{\log n}$, this in total gives a bound of the form $\frac{n}{2\sqrt{\log n}2^{3\sqrt{\log n}}}$ for a 3-AP-free set in $[n] \cup \{0\}$, and then removing 0 from this set gives your desired construction.

8.5 *Prove that* $\mathrm{ex}(n, K_3, K_4 - e) \geq n^2 e^{-C\sqrt{\log n}}$ *for some* $C > 0$ *[1+].*

Consider the graph G_A we used for Roth's Theorem with A the large 3-AP-free set we constructed in the previous exercise. This graph has no $K_4 - e$ because A is 3-AP-free and its number of triangles is exactly exactly $|A|n$, proving the result.

10 HW 10

8.7a *Prove that if* (V_i, V_j) *is an* ε*-regular pair with* $|V_i|, |V_j| \geq \frac{1}{2}M(\varepsilon)^{-1}n$ *and* $d(V_i, V_j) \geq \frac{1}{2} + 2\varepsilon$ *and if* $\alpha(G) < \frac{1}{4}\varepsilon M(\varepsilon)^{-1}n$*, then* G *contains a* K_4 *(Hint: why can't* G *just be a complete bipartite graph on* $V_i \cup V_j$*?) [2].*

Let $V_i' \subseteq V_i$ be the vertices with degree at least $(\frac{1}{2}+\varepsilon)|V_j|$ into V_j. Because we're ε-regular we must have $|V_i'| \geq (1-\varepsilon)|V_i| \geq (1-\varepsilon)\frac{1}{2}M(\varepsilon)^{-1}n > \alpha(G)$ so there's some edge xy in V_i'. Now the common neighborhood of x, y into V_j has size at least $2\varepsilon|V_j| \geq \varepsilon M(\varepsilon)^{-1}n \geq \alpha(G)$ so again we can find an edge uv in V_j which together with x, y forms a K_4, a contradiction.

8.7b *Let* V_i, V_j, V_k *be three (possibly non-distinct) sets in the partition such that all three of their pairs are* ε*-regular. Prove that if all three of these sets have size at least* $\frac{1}{2}M(\varepsilon)^{-1}n$*, if all three of their pairs have density at least* 2ε*, and if* $\alpha(G) < \frac{1}{2}\varepsilon^2 M(\varepsilon)^{-1}n$*, then* G *contains a* K_4 *[2].*

Let $V_i' \subseteq V_i$ be the vertices with at least $\varepsilon|V_j|, \varepsilon|V_k|$ neighbors in $|V_j|, |V_k|$. Again we have $|V_i'| \geq (1-2\varepsilon)|V_i| > 0$; pick any vertex x_i in there and let V_j', V_k' be its neighborhoods. Now let $V_j'' \subseteq V_j'$ be the vertices with at least $\varepsilon|V_k'|$ neighbors in $|V_k'|$. The number of vertices in V_j' which fail to have this property is less than εn since $|V_k'| \geq \varepsilon|V_k|$, so there exists some $x_j \in V_j''$. Now let V_k'' be the neighborhood of x_j in V_k' (which itself is in the neighborhood of x_i). Because $|V_k''| \geq \varepsilon^2|V_k|$ there exists an edge uv in this set which together with x_i, x_j forms a K_4.

8.7d *Put these pieces together to prove the following fact: for all* $\varepsilon' > 0$ *there exists some* $\varepsilon > 0$ *such that every* n*-vertex* K_4*-free graph* G *with* $\alpha(G) < \frac{1}{2}\varepsilon^2 M(\varepsilon)^{-1}n$ *satisfies*

$$e(G) \leq \left(\frac{1}{8} + \varepsilon'\right) n^2.$$

Take an equitable ε-regular partition and delete edges involved with irregular pairs and with $d(V_i, V_j) < 2\varepsilon$. After doing this we still have most of our edges. Consider all the pairs (i, j) such that there is an edge between V_i and V_j (meaning the pair is ε-regular with density at least 2ε). We must have $i \neq j$ for all such pairs by (c). Also observe by (a) that the number of edges which are between these two pairs is at most

$(\frac{1}{2}+2\varepsilon)|V_i||V_j| \approx (\frac{1}{2}+\varepsilon)(n/m)^2$. Because the total number of edges is at least $(1/8+\varepsilon'/2)n^2$ this is only possible if the number of such pairs (i,j) is at least $(\frac{1}{4}+\varepsilon''')m^2$. But now if we define a graph on these parts where we have an edge if we have one of these pairs (i,j) then by Mantel's Theorem there must exist a triangle, which by (b) gives a copy of K_4.

11 HW 11

9.2 *Prove that paths of the form P_{2^r+1} for some integer $r \geq 0$ are Sidorenko [2].*

Let $d(v)$ denote the number of homomorphisms of $P_{2^{r-1}+1}$ there are with v as an endpoint, then $|Hom(P_{2^r+1})| = \sum d(v)^2$ and now convexity and induction gives it to you.

9.3b *Prove that if F is a graph and if there exists some $c > 0$ such that $t_F(G) \geq ct_{K_2}(G)^{e(F)}$ for all graphs G, then F is Sidorenko [2+].*

Assume for contradiction that there exists a G with $t_F(G) = (1-\varepsilon)t_{K_2}(G)^{e(F)}$ for some $\varepsilon > 0$. Let G^n denote the graph obtained by taking the tensor product with itself n times. By hypothesis and the previous part we have $ct_{K_2}(G^n)^{e(F)} \leq t_F(G^n) = t_F(G)^n = (1-\varepsilon)^n t_{K_2}(G)^{ne(F)} = (1-\varepsilon)^n t_{K_2}(G^n)^{e(F)}$, and this a contradiction for n sufficiently large in terms of c, ε.

9.4b *Prove that a bipartite graph is Sidorenko if and only if it satisfies the Erdős-Simonovits Supersaturation Conjecture II. [2]*

Take an arbitrary graph G. For large n, the blowup $G[n]$ will be dense enough to apply Erdos-Simonovits to get many copies of F, this in turn gives many homomorphisms in G. In particular, we have a lower bound of the form $ct_{K_2}(G)^{e(F)}$ so we win by the previous exercise.

12 HW 12

9.8 *Prove that if F is not edge-critical then $\mathrm{ex}(n,F) \geq e(T_{\chi(F)-1}(n)) + 1$ whenever $n \geq \chi(F)$. Further, prove that if $\mathcal{M}_F$ is the set of bipartite graphs F' which can be obtained by taking a $\chi(F)$-proper coloring of F and then deleting $\chi(F)-2$ of its color classes, then*

$$\mathrm{ex}(n,F) \geq e(T_{\chi(F)-1}(n)) + \mathrm{ex}\left(\left\lceil \frac{n}{\chi(F)-1} \right\rceil, \mathcal{M}_F\right)$$

[2-].

For the first part take a Turán graph and an edge within the largest part. For the second, add an $\mathcal{M}_F$-free graph within the largest part.

11.3 *Prove that if an (n,r,k,d)-Steiner system exists then $\binom{r-i}{k-i}$ divides $d\binom{n-i}{k-i}$ for all $0 \leq i \leq k$.*

Fix an arbitrary set $S \subseteq V(H)$ of size i and count all the pairs (K,e) where K is a k-set containing S and e is an edge containing K. Note that the total number of choices of K

is exactly $\binom{n-i}{k-i}$ and given this the number of edges is exactly d. On the other hand, the number of choices for e is $\deg(S)$ and each of these edges e appear in $\binom{r-i}{k-i}$ pairs. Thus, $d\binom{n-i}{k-i} = \binom{r-i}{k-i}\deg(S)$ which gives the divisibility condition.

11.4a *Prove that* $\mathrm{ex}(n, C_\ell^{(3)}) = \Theta(n^2)$ *for all* $\ell \geq 3$ *[2].*

Assume H is an n-vertex 3-graph with, say, $4\ell\binom{n}{2}$ edges. Iteratively go through each pair of H with degree less than 3ℓ and remove all edges containing the pair, and let H' be the resulting 3-graph. Note that $e(H') \geq e(H) - 3\ell\binom{n}{2} > 0$, so we can let $\{x_1, x_2, x_\ell\}$ be an arbitrary edge. Iteratively given the edge $\{x_i, x_{i+1}, x_\ell\}$ we let $\{x_{i+1}, x_{i+2}, x_\ell\}$ be any edge of H' which contains x_{i+1}, x_ℓ where $x_{i+2} \neq x_j$ for any $j \leq i+1$, noting that we can do this because every pair in H' with positive degree has large degree. This implies there exist edges $\{x_i, x_{i+1}, x_\ell\}$ for all $1 \leq i \leq \ell - 2$. In particular, for all $1 \leq i \leq \ell$ there exists an edge of H' containing the pair x_i, x_{i+1} with indices written modulo ℓ. We can thus find for each pair an edge $\{x_i, y_{i,i+1}, x_{i+1}\}$ with the $y_{i,i+1}$ distinct from all previously selected vertices, giving a copy of $C_\ell^{(3)}$.